ANDINO: Automata Network Model for the Diffusion of Nitric Oxide in Biological and Artificial Neural Systems. A Preliminary Study

Suárez-Araujo C.P., Fernández-López P., and García-Báez P.

Abstract— **An underlying mechanism of Volume Transmission (VT) is the diffusion of Nitric Oxide (NO), which affects all types of brain activity. In this paper we present a new discrete Model based on Automata Networks for Diffusion of NO (ANDINO). The main objectives are to demonstrate its ability to observe the dynamics of diffusion of NO from a molecular perspective and to study how the establishment of local diffusion rules produces global phenomena in which complex structures of NO molecules are generated,(following specified movement dynamics). We show that ANDINO is a highly powerful and general formal tool for studying and determining the dynamic of NO diffusion, both in brain and artificial neural systems. We present a short (and preliminary) computational analysis of the dynamics of NO diffusion and the self-regulation process using a two-dimensional AN made up of 420 automata with a hexagonal connection scheme working in a homogeneous and isotropic environment. Finally we demonstrate the ability of ANDINO to analyze the relationship among the different processes involved in the global dynamics of NO, such as diffusion, self-regulation/recombination and Synthesis.**

I. INTRODUCTION

Brain activity is induced by the functional coupling of many nerve cells, cellular communication and learning [1]. These aspects involve biological and artificial neural networks (B/ANN) and depend on an underlying neural signaling scheme. Knowledge of these mechanisms allows for a better understanding of brain computational style, which is one of the biggest challenges in computational neuroscience. Furthermore, it would play a crucial role in potential ANN computation.

At present, a new type of neural signaling, called volume transmission (VT) seems to be emerging. Its underlying mechanism is the diffusion of neuroactive substances and diffusible signals, such as nitric oxide (NO), in the extracellular space (ECS), which affect brain activity in a global way [2], [3] and is considered an atypical cellular messenger.

NO is a liposoluble molecule with extreme diffusibility in both aqueous and lipid environments. This characteristic allows for a rapid three-dimensional spread of the signal irrespective of the presence of membranes [4]. This condition lets it freely diffuse through membranes affecting all neighbouring cells [5] and [6]. It can act as a retrograde neurotransmitter; involved in learning and memory, and it can play a role in the long term potentiation process (LTP).

The dynamics of NO is made up of several processes: Synthesis, Diffusion, and Self-Regulation/Recombination with other substances. This work is centered on the Diffusion and Self-Regulation processes.

The NO is synthesized by the NO synthase (NOS) from L-arginine-NO path [5], [7], [8] and [9]. The Diffusion of NO is governed by its concentration gradient and it does not need recapturing processes. As long as the diffusion occurs, Self-Regulation and Recombination with different substrates appear [10].

Several models from different perspectives have been developed in order to understand how NO functions as a neural signaling molecule due to their absence in determinant experimental data. These models include: Analytic models: studying the diffusion process [11], [12], even considering the morphology of the synthesis process [13], [14], driven to analysis of parameters of the diffusion process [15]. Discrete or Semi-Discrete Models oriented to the study of the diffusion process [11] and [16].

This paper presents a new discrete Model based on Finite Automata Networks for Diffusion of NO (ANDINO). Its main objective is to be able to observe the dynamics of diffusion of NO from a micro-perspective which assumes that the dynamics of every NO molecule are known. In addition the paper aims to study how the establishment of local diffusion rules produce global phenomena in which complex structures of NO molecules are generated, following a determined dynamics of movement. It will be shown, by means of computational studies, that ANDINO allows the diffusion of NO to be studied from its perspective under the transport phenomena, where the basic process is random molecular. We present a short computational analysis of the dynamics of diffusion of NO and the self-regulation process. We will show that ANDINO is a highly powerful and general formal tool for studying and determining the dynamics of diffusion of NO, both in brain and artificial neural systems. Finally, we will observe the capability of ANDINO to analyze the relation among the processes involved in the global dynamics of NO, diffusion, self-regulation and recombination and Synthesis.

II. AUTOMATA NETWORK MODEL FOR THE DIFFUSION OF NO (ANDINO)

The NO diffusion model under study in this paper is discrete, allowing the dynamic behavior of NO to be

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modeled from a microscopic point of view, at the molecular level. The model is a non cyclical bi-dimensional *Automata Network* (AN) [17] with a dynamic structure, where each automata represents a NO molecule. The connectivity structure is hexagonal, (see figure 1), and we can observe a maximum neighborhood of three neighbors for each automata. The number of neighbors by automata is related to the automata position in the AN, (see figure 1). Consideration of not only the number of automata that make up a neighborhood, but also the situation of each element inside the AN structure reveals different neighborhoods. Each automata in the AN is characterized and defined by its neighborhood, (see figure 1). Each automata identification is carried out by means of a naming convention that uses the nomenclature in geographical orientation. We can now identify the positions of the automata. The NW-NE automata, whose location is in the lower part of the AN, the SW-SE automata, which refers to those that are located in the upper part of the network, the NE-S, N-SE, NW-S and N-SW automata, which belong to the sides of the AN, and automata NW-NE-S and N-SW-SE automata, those that are found in the interior of the AN. In all of these cases N ≡ North, S ≡ South, NW ≡ Northwest, NE ≡ Northeast, SW ≡ Southwest and SE ≡ Southeast.

The characteristics of the dynamic structure that make up the AN diffusion model of NO (ANDINO) can be associated with the same dynamic of the diffusion and the self regulation of NO. This capacity provides the model with great flexibility and structural adaptability, resulting in great potential regarding the modeling of a phenomenon that contains important degrees of variability.

The phenomena of self-regulation creates structural changes in the AN because the disappearance of the NO produces the disconnection of automata in the designed network, which produces changes in the neighborhoods and in the connectivity topology, and affects the final structure of the automata.

Also note that the diffusion process from each automata in their displacements in the configuration space produces a reconnection, and also generates a change in the neighborhoods and in the automata structure.

The automata network model for NO diffusion requires a transition function. A phenomenological framework is established for the formalization needed of the NO diffusion process with the aim of modeling it. The objective is to model the dynamics of NO diffusion, focusing on the molecular movement in a homogeneous and isotropic environment, without considering the possible biochemical processes that it can be subjected to. This movement dynamic of the NO gas has its origins in the phenomenological laws of the transport of matter.

Our model studies the dynamics of the NO once the NO has been synthesized, that is, it focuses on the diffusion processes and self-regulation/recombination of the complete NO dynamics, figure 1. In the first of these processes, the NO molecules are freely displaced in the environment until they reach their destinations and recombine or self-regulate. The self-regulation of the NO occurs in one of two ways: 1) average life time or 2) avoiding situations characterized by toxicity because of high concentrations of NO in a specified area.

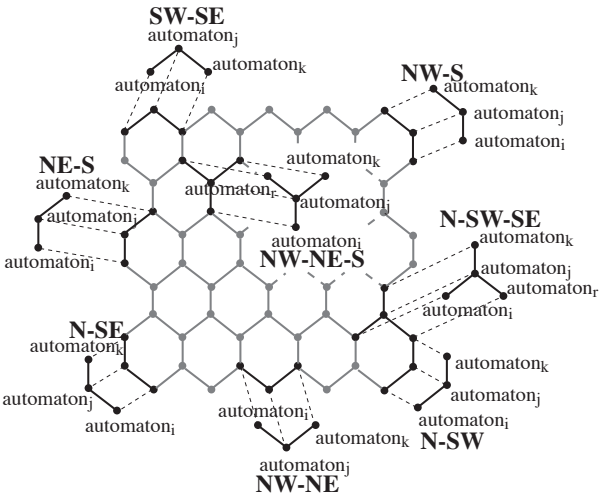


Fig. 1. Automata types as a function of its neighborhood, with two or three neighbors.

Using this phenomenological framework we establish the following states that the NO molecules must pass through in the diffusion process in biological and artificial environments:

Diffusion. NO molecules found in this state are displaced in the environment, and are only influenced by nearby molecular movements (neighboring molecules). In order to completely gather the diffusion dynamics, we define associated sub-states by molecular displacement orientation. By using geographic nomenclature we can identify 6 diffusion sub-states, for each automaton: N ≡ spread to the North, NE ≡ spread to the Northeast, SE ≡ spread to the Southeast, S ≡ spread to the South, SW ≡ spread to the Southwest, NW ≡ spread to the Northwest.

Self-regulation and Recombination. Both of these NO chemical processes form part of the control of specific aspects such as NO level achieved, time length of said effect, environment where the process occurs until its disappearance. We focus on the *Self-regulation* process and modeling the two possible causes that the NO molecules bring to the self-regulation state: A1 ≡ elapsed average life time and A2 ≡ saturation of NO levels. We create two situations with the other two states in which other automata of the designed AN can also be found.

Hence, based on these described states, we define the transition function of the automata. We propose two versions of this function based on two possible dynamics of NO molecular movement. The use of the proposed model allows us to analyze the implications that each one of them displays NO behavior and its effect on the biological and artificial settings where they take place.

Transition function A (TFA) represents the first version of the transition function, and gathers some behavior rules where movement dynamics of the NO molecules (for each network automata) indicate that the movement of each NO molecule goes in an opposite direction than its neighbors. Additionally, the second version of the defined transition function (*Transition Function B, TFB*), is used when the

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movement of the NO molecule stays in the same direction of the movement of its molecule neighbors.

We define these transition functions by using modular arithmetic and then establishing a correspondence between the set of states S = {N, NE, SE, S, SW, NW, A1, A2}, from the different possible automata states, and the next finite subset of integers V = {0, 1, 2, 3, 4, 5, -1, -2}.

TFA, as mentioned previously, is associated with a movement dynamic where the NO molecule follows movement steps which are opposite those of its neighbors. Once type automata$_j$ has been found in any of the substates related to the *Diffusion process*, {0, 1, 2, 3, 4, 5}, in time *n*, the automata$_j$ state in time *n+1* is calculated by following these TFA rules:

i. If automata$_j$ does not have a neighbor ⇒ continue in the same diffusion substate {N ≡ 0, NE ≡ 1, SE ≡ 2, S ≡ 3, SW ≡ 4, NW ≡ 5} until the average time that has been assigned to the automata passes, later continuing on to state A1 ≡ -1.

ii. If automata$_j$ has one neighbor ⇒ Calculate its new state based on automata$_i$ state, only neighbor from the first, according to the following rule: (automata$_i$ state + 3) mod 6.

iii. If the automata has two neighbors, than one of the following two types of automata occurs: NE-S, N-SE, NW-NE, SW-SE, NW-S, N-SW; the new state is obtained based on the following equation of the TFA:

Go to state A2 ≡ -2 (Self-regulation according to NO level saturation), if it satisfies some of the following conditions:

a. Automata$_j$ type is NW-NE ∧ automata$_i$ state = 2 ∧ automata$_k$ state = 4.
b. Automata$_j$ type is SW-SE ∧ automata state $_i$ = 1 ∧ automata$_k$ state = 5.
c. Automata$_j$ type is NE-S ∧ automata$_i$ state = 4 ∧ automata$_k$ state = 0.
d. Automata$_j$ type is N-SE ∧ automata$_i$ state= 3 ∧ automata$_k$ state = 5.
e. Automata$_j$ type is NW-S ∧ Automata$_i$ state = 2 ∧ automata$_k$ state= 0.
f. Automata$_j$ type is N-SW ∧ Automata$_i$ state = 3 ∧ Automata$_k$ state = 1.

Movement to a state E ∈{0, 1, 2, 3, 4, 5} occurs if the following general condition is satisfied:

[Automata$_i$ state, Automata$_k$ state] ∈ {[(E + 3) mod 6, (E + 3) mod 6], [(E + 2) mod 6, (E + 4) mod 6], [(E + 4) mod 6, (E + 2) mod 6], [(E + 3) mod 6, (E + 4) mod 6], [(E + 4) mod 6, (E + 3) mod 6]}. C1

iv. If the automata has three neighbors (automata type NW-NE-S and N-SW-SE), we calculate the new state according to the following:

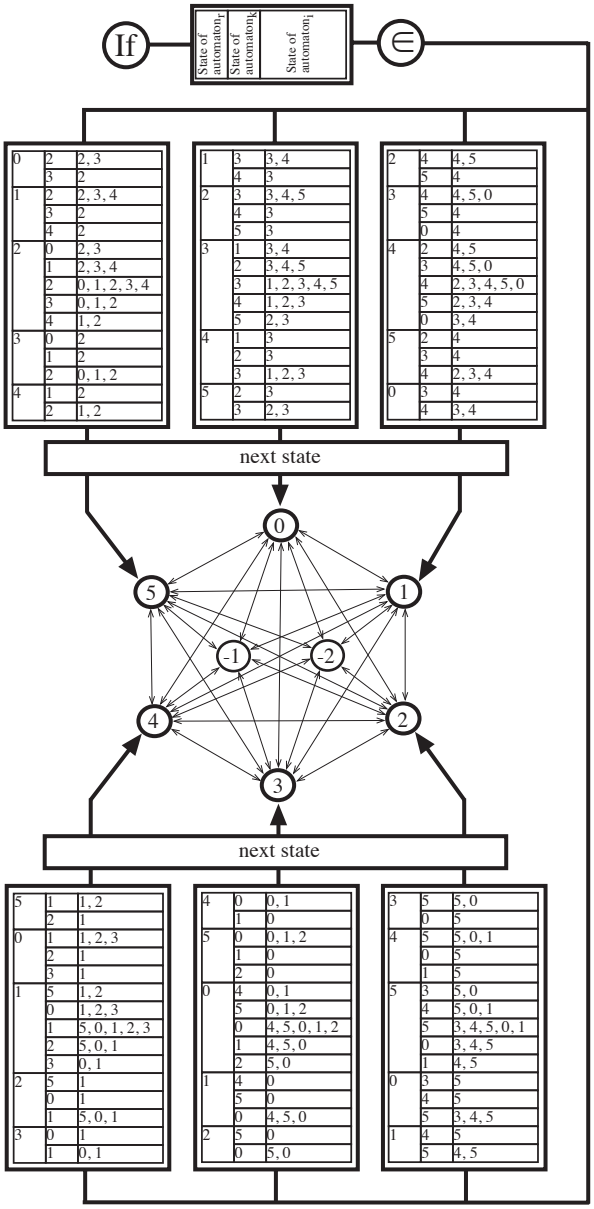


Fig. 2. Set of possible neighboring states of an automata whose neighborhood is made up of three automata.

The automaton goes to state A2 ≡ -2, if it satisfies some of the following conditions:

a. Automata$_j$ type is NW-NE-S ∧ automata$_r$ = state 2 ∧ automata$_k$ state = 4 ∧ automata$_i$ state = 0.
b. Automata$_i$ type is N-SW-SE ∧ automata$_r$ state = 3 ∧ automata$_k$ state = 1 ∧ automata$_i$ state = 5.

The automata moves to state E ∈{0, 1, 2, 3, 4, 5} if the following general condition is satisfied:

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$$[\text{Automata}_r \text{ state}, \text{Automata}_k \text{ state}, \text{Automata}_i \text{ state}] \in \{[(E+1) \bmod 6, (E+3) \bmod 6, (E+3) \bmod 6], [(E+1) \bmod 6, (E+3) \bmod 6, (E+4) \bmod 6], [(E+1) \bmod 6, (E+4) \bmod 6, (E+3) \bmod 6], \ldots, [(E+5) \bmod 6, (E+3) \bmod 6, (E+3) \bmod 6]\}, \quad \text{C2}$$

where, in figure 2, we can observe all of the possibilities from C2.

Note that in rule iii and in rule iv we assume that the lifetime of the automata has not elapsed and that we can proceed to self-regulation state A1 when it has also elapsed.

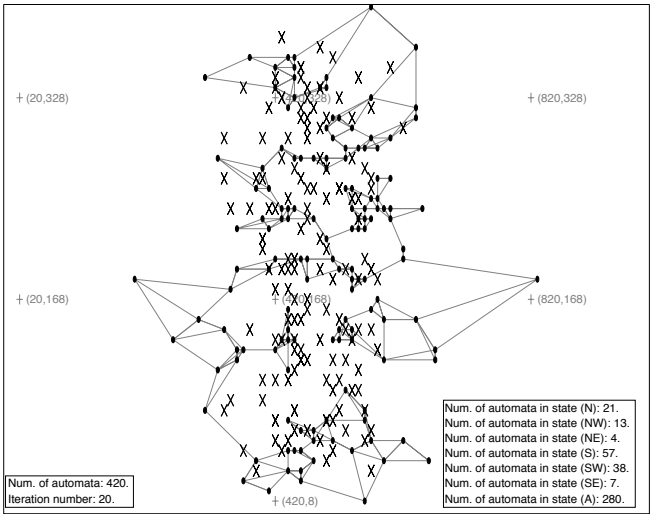
TFB is associated with a movement dynamic where each NO molecule follows the same steps of movement as their neighboring molecules. Its definition contains the same structure as the definition of the TFA, except the way in which the change occurs in the automata state in rule ii, and in rules iii and iv, conditions C1 and C2 have different values, so that the way these changes in the states are produced in the automata_j, following the states that neighboring automata display.

III. RESULTS AND DISCUSSION

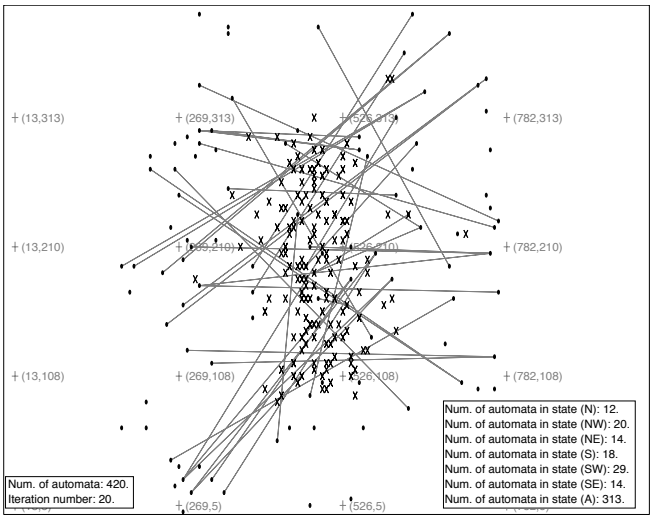
Diffusion is the main axis in the study of NO dynamics [4]. However, diffusion is also affected by another important part of the NO dynamics, the self-regulation process. Diffusion can be considered as the agent that is most responsible for the influence of NO on different brain zones from a functional and structural point of view.

We have performed the model analysis by studying the dynamics of NO using a two-dimensional AN constituted by 420 automata with a hexagonal connection scheme working in a homogeneous and isotropic environment. The AN is set out in a rectangular shape in the center of the environment, with an initial configuration of 66 automata in the N state, 81 in the NW state, 61 in the NE state, 71 in the S state, 68 in the SW state, 73 in the SE state and 0 in the self-regulation state. The AN evolves with time, until 101 iterations are available for this study, according to their transition functions.

The proposed model contains a feature which allows dynamic or fixed connection topology, thus it is embodied in the type of neighborhood found in the automata network. The first type of this AN evolves within time and is modified along the neighborhood of the automata in each iteration, re-adjusting itself to the closest automata. This structural change causes a continuous readjustment of local information in space and time, which influences how the NO molecular movement is guided without information on the process of synthesis. The fixed connection model has static automata neighborhoods. Each automata has a set of neighbors from the onset, and this group is maintained throughout the evolution of the molecular movement, as long as their last states do not enter into a self-regulation state. In this case, the neighborhood then changes, since the automata in self-regulation will disappear in the neighborhood. In this behavior mode we observe that the molecular movement of NO is conditioned by the movements of the neighboring molecules from the process of synthesis. It is more than likely that the NO molecular dynamics are influenced by the dynamics of those molecules that are nearby, and not by the ones that were nearby in the process of synthesis. Nevertheless, it is possible that these latter dynamics did in some way influence the diffusion.



a)



b)

Fig. 3. Temporal evolution of ANDINO with transition function TFA, a) dynamic neighborhood, and b) static neighborhood.

It will be seen that this AN-based model allows us to study the behavior of the NO from a microscopic to a macroscopic level. It is an appropriate framework to collect any impressions on the dynamics of NO Diffusion or any local rule of the same dynamics, permitting observations of the global behavior of the NO that it generates. In addition, it allows us to analyze structural and/or functional aspects of the NO mechanism, accepting the hypothesis and/or events as well as create new hypotheses on the same conditions.

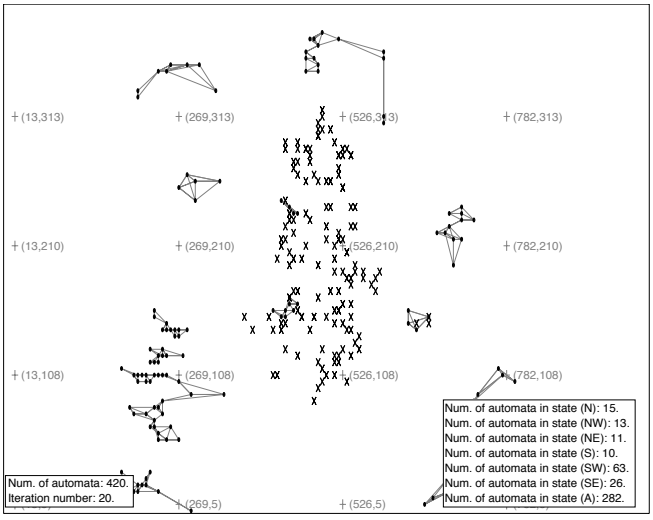
In the first part of the study, the dynamics of the NO diffusion will be analyzed, studying the behavior of the model with its defined configuration and the two defined transition functions, TFA and TFB.

The behavior of the model using transition function TFA is shown in the figure 3, and allows us to see the evolution

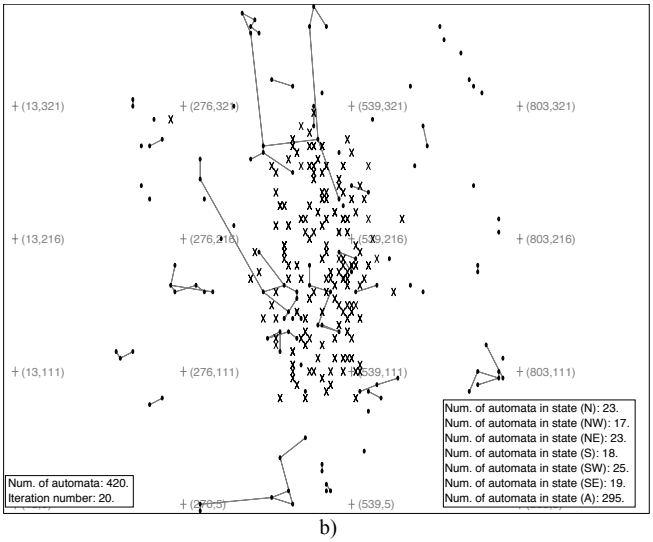
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of movements that the NO molecules follow after 30 iterations.

Both cases reveal that a large number of NO molecules suffer self-regulation by NO level (cross marks) close to the process of synthesis. This fact is in agreement with the work hypothesis that has been established based on biological data. NO self-regulation is greater when the AN is made up of automata with dynamic neighborhoods. This behavior happens because schemes that are void of reconnection show greater expansion of molecules than those where reconnection is present. In addition, in this last scheme, the defined rules at a local level based on TFA seem to limit the expansion of these molecules at a global level. These considerations allow us to conclude that the self-regulation by NO level in our model has a local character, reaching maximum levels in the initial stage of the diffusion process, exactly in the zone of the synthesis.



a)



b)

Fig. 4. Temporal evolution of ANDINO with transition function TFB, , a) dynamic neighborhood, and b) static neighborhood.

With regard to the possible presence of patterns in the dynamics of NO diffusion, figures 3a and 3b show that the use of dynamic neighborhoods does not induce the formation of clusters with independent displacement, while the static neighborhoods allow us to observe the formation of molecular groupings, which, although not having an independent dynamics, do show joint displacement.

NO behavior for TFB, figure 4, is seen as the opposite of what happens with TFA. Figure 4 show the situation at the same iteration of the NO dynamics as in the figure 3.

A high level of self-regulation for NO levels is observed in TFB, slightly lower than that one found in TFA, specifically in situations close to the process of synthesis. Even though the limits of diffusion are similar, the expansion of NO molecules in the scheme of dynamic neighborhoods (figure 4a) is greater than in static neighborhoods (figure 4b). With respect to local features of the self-regulation process by NO level in our model, we have a scheme which reflects greater capacity for expansion that will have a smaller number of molecules (automata) that experience self-regulation by NO level and, consequently, a greater number of automata in a diffusion state.

Both of the analyzed dynamics show molecular grouping at generation number 20. In the first scheme, with dynamic neighborhoods, groupings have independent behavior and movement. On the other hand, the second scheme, using static neighborhoods, show molecular groupings that are displaced jointly, with a dependent type of dynamics.

Thus, it can be inferred that the formation of molecular groupings depends, in some way, on the capacity of NO molecules to expand. Thus, expansion can be considered as an underlying aspect of the NO diffusion, and based on its global behavior, which is perceived to be random, is the product of the establishment of local, deterministic type rules.

On the other hand, the independent character, not only in behavior but also in displacement, of the groupings of molecules depends on reconnection. Hence, it is in the dynamic neighborhood schemes where behavior of the NO molecule is subject to local information; where more groupings and complex structures are produced that arrive from different parts of the environment during its task of transmitting pieces of information.

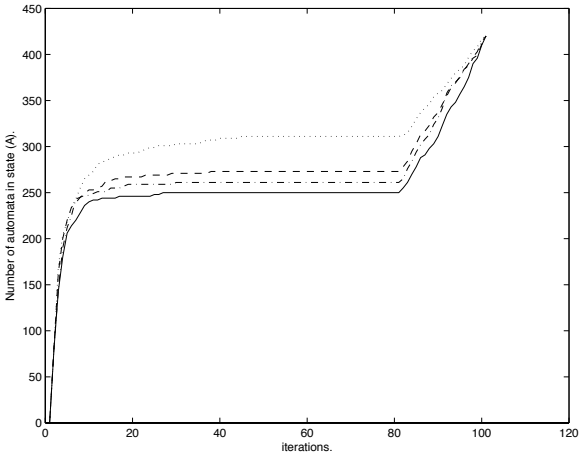


Fig. 5. Temporal evolution of automata in self-regulation state under four scenarios: TFA and a dynamic neighborhood scheme [-], TFA and a static neighborhood scheme [–.], TFB and a dynamic neighborhood scheme, [..], TFB and a static neighborhood scheme [--].

Considering these facts, we can conclude that the formation of molecular groupings, with displacement and independent behavior, is better under the transition function

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TFB and a dynamic neighborhood structure. Observed from an information transmission perspective, this structure-dynamic combination reflects the most selective element in this transfer, opposite of what occurs in a more generalized transmission and in diffusion.

We finish the analysis of the model with the presentation (figure 5) of temporal evolution of the automata in the self-regulation states in all of the scenarios that we studied in our proposed model and where we were able to observe a first stage, close to the process of synthesis, where self-regulation by NO level takes place in more than 50% of the molecules, a second more stable stage related to the self-regulation of the NO level and a third, shorter stage, where self-regulation begins due to elapsed average lifetime of the NO. Note that the four scenarios in this study in the TFB with dynamic neighborhoods are higher in the first stage, which lets us conclude that when NO molecules show a movement dynamics that follows its neighbors and these in turn change throughout the time, the capability of expansion is minimal and, consequently, the NO level reaches its highest values, generating important levels of self-regulation by NO level. According to our model this happens close to the process of synthesis.

IV. CONCLUSIONS

This study presents and analyzes an automata network model for the diffusion of NO and demonstrates the model capability to study and emulate the dynamics of the diffusion of NO as the underlying molecule in the mechanisms for signaling in the brain and in artificial neural systems.

The model reveals an ability to show how the introduction of local rules that define the molecular movement of the NO induces global behaviors from the NO. It facilitates the visualization of the NO behavior. ANDINO is defined from the substance perspective, and it is a specific model, as opposed to the abstract nature of those models that have been developed up to now. These models, including behavioral models, are defined from the perspective of a perceived variable from the substance, that is, it models a representation of itself.

We consider different diffusion scenarios of the NO based on specific parameterizations of the transition function and the type of neighborhood. Two classes of transition functions have been proposed that gather possible behaviors of the molecular movement of NO (TFA and TFB) and two types of neighborhoods, a dynamic neighborhood, based on closeness criteria, and static neighborhoods.

In our model we gather information on two types of self-regulation: a) by level of the NO and b) self-regulation at the end of average life of NO. The first one is essentially located very close to the process of synthesis. The self-regulation process seems to be independent with respect to TFA and TFB.

The dynamic neighborhood generates a greater expansion of the substance, and permits the generation of complex structures to improve based on the grouping of molecules with autonomous (independent) behaviors and displacements. Additionally, in the schemes where the molecular behavior of NO depends on local information, and where the movement of a molecule follows the travel of local neighbors, an improved grouping of molecules is created with more powerful complex structures. These structures reach to different parts of the environment in order to transmit information selectively.

On the other hand, we have observed that the self-regulation by the term of the average life time of the NO reveals a linear relationship, until the end of the diffusion process, and has a direct impact on the final reach of the NO in the environment.

Future research will focus on the acquisition of a detailed formalization of the processes that are involved in the dynamics of the NO, where the perspective from the analysis of attractors, cycles and paths of the molecules and the initial configuration of the automata network are all considered.

REFERENCES

[1] Suárez Araujo, C.P., "Study and Reflections on the Functional and Organisational Role of Neuromessenger Nitric Oxide in Learning: An Artificial and Biological Approach," *Computer Anticipatory Systems*, AIP 517, 2000, pp. 296-307.

[2] Syková, E. "The extracellular space in the CNS: its regulation, volume and geometry in normal and pathological neural function," *The Neuroscientist* 3,1997, pp. 28-41.

[3] Syková, E. "Extrasynaptic Volume Transmission and Diffusion parameters of the Extracellular Space," *Neuroscience* 129, 2004, pp. 861-876.

[4] Suárez Araujo, C.P., Fernández López, P. and García Báez P., "Nitric Oxide Diffusion Attributes in Biological and Artificial Environments: A Computational Study," *Majlesi Journal of Electrical Engineering* Vol. 5, No. 1, 2011, submitted for publication

[5] Schuman, E.M. and Madison, D.V. "Nitric Oxide and Synaptic Function," *Annu. Rev Neurosci.* 17, 1994, pp. 153-83.

[6] Garthwaite, J. and Boulton, C.L. "Nitric Oxide signalling in the Central Nervous System," *Annu. Rev. Physiol.* 57, 1995, pp. 683-706.

[7] Huizhong W. and Mu-ming Poo. "Retrograde signalling at central synapse," *PNAS*, vol 98, no 20, 2001, pp. 11009-11015.

[8] Bredt, D.S. and Snyder, S.H. "NITRIC OXIDE: A Physiologic Messenger Molecule," *Annu. Rev. Biochem.* 63, 1994, pp. 175-95.

[9] Moncada, S. Et al. "The L-Arginine: Nitric oxide pathway," *J. Cardiovasc. Pharmacol.* 17, 1999, S1-S9.

[10] Regidor, J. and Suárez Araujo, C.P. "Retrograde neural Menssenger in the Brain. Preliminary Study on the Implications in the Artificial Neural Networks," *Brain Processes, theories and Models. MIT Pres,* 1995, pp. 360-369.

[11] Lancaster, J.R (JR.). "Simulation of the diffusion and reaction of endogenously produced nitric oxide," *Proc. Natl. Acad. Sci. USA*, Vol. 91, 1994, pp. 8137-8141.

[12] Wood, J. and Garthwaite, J. "Models of the Diffusional Spread of Nitric Oxide: Implications for Neural Nitric Oxide Signalling and its Pharmacological Properties," *Neuropharmacology* Vol. 33, N. 11, 1994, pp. 1235-1244.

[13] Philippides, A, Husbands, P. And O'shea M. "Four-Dimensional Neuronal Signaling by Nitric Oxide: A Computational Analysis," *The Journal of Neuroscience*, 20(3), 2000, pp. 1199-1207.

[14] Suárez Araujo, C.P., Fernández López, P. and García Báez, P. "Towards a Model of Volume Transmission in Biological and Artificial Neural Networks: A CAST Approach", R. Moreno-Díaz et al. (Eds.): EUROCAST 2001, LNCS 2178, pp. 328-342.

[15] Vaughn, M.W., Kuo, L., and Liao J.C. "Effective diffusion distance of nitric oxide in the microcirculation," *Am J Physiol*, 1998, pp. 1705-1714.

[16] Suárez Araujo, C.P., Fernández López P.; García Báez P. and Simoes da Fonseca J.L., "A Model of Nitric Oxide Diffusion Based in Compartmental Systems," *In International Journal of Computing Anticipatory Systems (IJCAS)*, Vol. 18, 2006, pp:172-186.

[17] Dömösi, P. and Nehaniv C.L., "Algebraic Theory of Automata Networks", ISBN: 9780898715699, 2005.

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